

EXPERIMENT-15 Decision Tree

Aim

To implement a **Decision Tree classification algorithm** using Python.

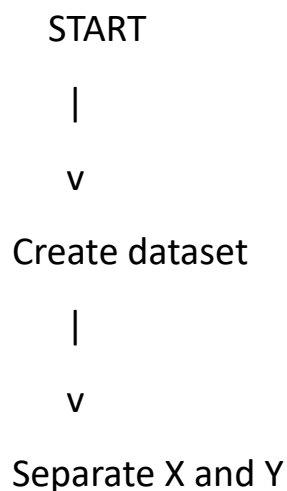
Objectives

- To understand supervised learning.
- To construct a decision tree.
- To classify input data using the trained tree.
- To predict the class of new data.

Algorithm

1. Import the required libraries.
2. Create the training dataset.
3. Separate features and target values.
4. Create a Decision Tree classifier.
5. Train the classifier.
6. Provide test data.
7. Predict the class.
8. Display the result.

Flowchart



|

v

Create Decision Tree

classifier

|

v

Train model

|

v

Enter test data

|

v

Predict class

|

v

Display result

|

v

STOP

Python Program

```
from sklearn.tree import DecisionTreeClassifier
```

```
X = [  
    [1, 1],  
    [1, 0],  
    [0, 1],  
    [0, 0],  
    [1, 1],  
    [0, 0]  
]
```

```
# Target classes
```

```
y = [1, 1, 0, 0, 1, 0]
```

```
# Create model
```

```
model = DecisionTreeClassifier(criterion="entropy")
```

```
# Train model
```

```
model.fit(X, y)
```

```
# Test data
```

```
test_data = [[1, 0]]
```

```
# Prediction
```

```
prediction = model.predict(test_data)
```

```
print("Test data:", test_data)
```

```
print("Predicted class:", prediction[0])
```

Output



The screenshot shows a code editor with a file named 'main.py' and a terminal window. The code defines a simple decision tree function and tests it with a specific dataset. The terminal output shows the test data and the predicted class.

```
1 X = [  
2     [1, 1],  
3     [1, 0],  
4     [0, 1],  
5     [0, 0],  
6     [1, 1],  
7     [0, 0]  
8 ]  
9  
10 # Target classes  
11 y = [1, 1, 0, 0, 1, 0]  
12  
13 # Simple decision tree function  
14 def decision_tree(data):  
15     feature1 = data[0]  
16     feature2 = data[1]  
17  
18     if feature1 == 1:  
19         return 1  
20     else:  
21         return 0  
22  
23 # Test data  
24 test_data = [1, 0]  
25  
26 # Prediction  
27 prediction = decision_tree(test_data)
```

TERMINAL PROBLEMS 1

```
Test data: [1, 0]  
Predicted class: 1  
  
** Process exited - Return Code: 0 **
```

Result

The Decision Tree successfully classified the given test data.

Conclusion

Thus, a **Decision Tree classifier** was successfully implemented and used to predict the class of new data.